

Article

Intelligent Unmanned Aerial Vehicle Swarm Control Under Electronic Warfare: A Cognitive–Intent Dual-Stream Reinforcement Learning Framework

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Highlights

What are the main findings?

- The Cognitive–Intent Decoupled Architecture (CIDA) resolves awareness loss in contested environments by separating belief reconstruction from tactical guidance.
- CIDA achieves a 96.15% mission success rate, significantly outperforming standard memory-augmented MARL baselines.

What are the implications of the main findings?

- Separating ‘where it is safe’ from ‘where to go’ avoids passive survival and brittle reward shaping in RF-denied swarms.
- The proposed steady-state mechanism stabilizes Transformer-based reinforcement learning, providing a reliable pipeline for safety-critical swarm deployments.

Abstract

Multi-unmanned aerial vehicle (UAV) platforms integrate radio-frequency (RF) sensing, datalinks, and onboard embedded compute; adversarial electronic warfare (EW) degrades these subsystems through jamming and forces decentralized control policies to act on fragmented observations—a setting aligned with intelligent electronic systems and autonomous robotics in contested spectrum. Cooperative swarms then face two compounding failure modes: loss of coherent situational awareness, and reward-driven passive survival that suppresses mission completion. Memory-based multi-agent reinforcement learning (MARL) partially addresses the first but tends to reinforce the second; dense intent shaping addresses the second but becomes unreliable when observations are incomplete. We propose CIDA (Cognitive–Intent Dual-Stream Architecture), a reinforcement learning framework that decouples belief reconstruction from tactical intent at the representation level while coupling them through a unified actor–critic update. The cognitive stream encodes a 64-step observation history with a pre-normalized Transformer to reconstruct threat belief; the intent stream supplies a hierarchical potential field (reconnaissance, threat-weighted engagement, and approach incentives). A steady-state training mechanism (dynamic reward scaling and adaptive gradient clipping) stabilizes Transformer-based on-policy learning under non-stationary multi-agent dynamics. In a complex terrain scenario with SAM, AAA, and jammer assets, CIDA reaches 96.15% task success versus 12.21% (memoryless PPO) and 25.28% (MAPPO+RNN), with ablations showing nonlinear coupling and emergent tactics such as jammer bypass and weak-sector traversal. Results are robust to a four-fold sweep of the intent-shaping weight (above 90% success).



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Keywords: UAV swarm; intelligent control; embedded autonomy; electronic warfare; jamming; multi-agent reinforcement learning; partial observability; Transformer; intent field; reward shaping; coordinated traversal

1. Introduction

The operational appeal of UAV swarms lies in their ability to distribute function and tolerate individual agent failure in ways that single platforms cannot [1–4]. From an *electronics and intelligent systems* perspective, each agent depends on onboard sensors, RF links, and embedded flight control; the vulnerability of UAV networks to interference and topology effects has been studied in the *Electronics* literature [5,6], complementing wireless-communications analyses of jamming in swarm links [7]. Against layered multi-tier threat environments employing dedicated electronic warfare, this distributed character becomes a liability: jamming cuts beyond-line-of-sight links, and no single agent has a complete picture of the operational area. The result is a fog-of-war regime in which the swarm's observe–orient–decide–act (OODA) loop [8] breaks at the sensing and communication stages, leaving agents to act on locally coherent but globally inconsistent environment state estimates [9]. Teaching a swarm to navigate and complete mission objectives under these conditions is the central problem of this paper.

Memory-augmented MARL is the natural response to fragmented perception [10–13]. Recurrent and attention-based policies can integrate observation sequences to infer hidden state, improving decisions under occlusion and communication loss [14,15]. The problem is that temporal memory alone does not give the swarm a reason to push forward. Trained with sparse extrinsic rewards, memory-equipped agents become skilled at avoiding threats but struggle to resolve the credit-assignment problem of deciding *when* to accept risk and advance. The result is a characteristic failure mode: high survivability and negligible mission success, with agents loitering at the edge of threat zones rather than advancing through them [16–19]. Compounding this, Transformer-based sequence models are particularly prone to abrupt policy collapse in non-stationary environments, raising reliability concerns for safety-critical deployment [20].

Dense reward shaping and potential fields attack the motivation problem directly [21,22]. By providing gradient information throughout the episode—not only at mission endpoints—potential fields can dramatically accelerate the emergence of goal-directed behavior [23,24]. Under jamming, however, field gradients computed from stale or incomplete observations can pull agents into defended sectors before they have sufficient situational awareness to navigate them safely, leading to unnecessary losses and myopic aggressiveness [25–27]. The target scenario studied here captures both failure modes simultaneously (Figure 1): five UAVs must traverse a complex terrain with four SAM sites, two AAA batteries, and a jammer radar that intermittently denies swarm communication and then neutralize the target facility.

We argue that survivability cognition—knowing where it is safe—and tactical intent—knowing where to go—are orthogonal concerns that should be handled by dedicated components and then coupled, not conflated within a single reward signal. This distinction drives our design. CIDA maintains a *cognitive stream* that reconstructs a threat belief from the agent's local observation history and an *intent stream* that translates mission priorities into dense reward gradients. The cognitive stream answers *is it safe to proceed here?*; the intent stream answers *which direction advances the mission?*. Neither interferes with the other during optimization, yet both feed the same actor–critic policy. A steady-state training mechanism—pre-normalization, dynamic reward scaling, and adaptive gradient clipping—

makes the long-horizon Transformer training reliable in this non-stationary multi-agent setting [6,7,28].

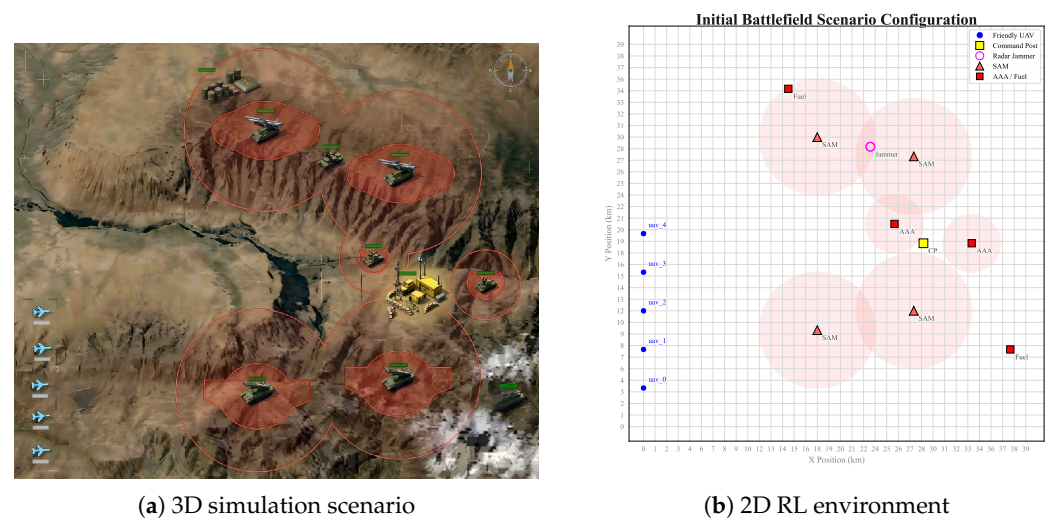


Figure 1. (a) The 3D hypothetical canyon terrain constructed from a simulation platform, showing terrain features and adversarial threat entities within their engagement zones (red-shaded regions). Cooperative UAVs are shown on the left in blue. (b) The corresponding 2D reinforcement learning environment, with cooperative UAVs initialized on the left and adversarial entities (SAM sites, AAA batteries, jammer radar, target facility (command post), and decoy fuel depots) distributed across the simulation area; circles denote engagement zones.

This paper makes the following algorithmic contributions, evaluated on a purpose-built simulation benchmark:

- A cognitive–intent dual-stream reinforcement learning framework for cooperative UAV swarm control, realized through two dedicated streams: a cognitive stream (Transformer-based belief reconstruction) and an intent stream (hierarchical potential field), enabling proactive mission completion without sacrificing survivability.
- A steady-state training mechanism—pre-normalization, dynamic reward scaling, and adaptive gradient clipping—that directly resolves the gradient-divergence instability intrinsic to Transformer-based on-policy RL in non-stationary multi-agent settings, a failure mode that standard PPO clipping cannot prevent on its own.
- Comprehensive simulation experiments showing CIDA achieves 96.15% task success in a fog-of-war complex terrain scenario, versus 12.21% (PPO) and 25.28% (MAPPO+RNN), with ablations confirming a nonlinear coupling gain and revealing qualitatively distinct emergent behaviors.

2. Related Work

2.1. MARL Under Partial Observability

Partial observability in cooperative UAV tasks is commonly formalized as a decentralized POMDP [29]. Recurrent policies extend standard MARL by maintaining a hidden state across steps, improving tracking and pursuit under occlusion [10,12,14]. Recent work has also explored deep deterministic policy gradient and multi-agent variants of actor–critic methods for UAV swarm task planning and cooperative decision-making in complex mission scenarios [3,4]. Transformer-based agents push this further by modeling long-range temporal dependencies and have shown clear representation gains in sequential control benchmarks [11,15,30]. Structured communication—through coordination graphs and bandwidth-constrained message encoding—has also been explored to share partial observations under network constraints [31–33]. Despite these

advances, a consistent empirical finding is that sequence models trained with standard on-policy updates exhibit high variance and occasional collapse in highly dynamic environments [20], a problem that grows more acute as the history window lengthens. CIDA addresses this directly: rather than treating the Transformer purely as a feature extractor, we design the cognitive stream specifically around belief reconstruction and pair it with a steady-state mechanism that targets the instability mechanisms known to affect Transformer RL.

2.2. Reward Shaping and Potential Fields

Potential fields provide a principled way to encode spatial objectives as dense reward signals, accelerating learning in navigation and obstacle-avoidance tasks [21–23,34]. In multi-agent air combat, hierarchical shaping based on virtual force fields or magnetic-field analogies has been effective at inducing coordinated maneuver behavior [24,35]. The well-known limitation is that potential fields are defined over an observed state: under jamming, where observations are fragmented and intermittent, field gradients computed from stale data can pull agents into configurations that are locally attractive but globally hazardous. The standard fix—better sensors or communication—is unavailable in the electronic warfare settings we target. Instead, CIDA’s intent stream operates on top of the cognitive stream’s reconstructed belief; so, its gradients reflect both mission priorities and an internally consistent threat estimate.

2.3. Training Stability in Deep RL

PPO and its multi-agent variants are the de facto standard for cooperative control tasks, valued for their relative robustness and ease of tuning [36,37]. Large-scale empirical studies reveal, however, that on-policy stability is sensitive to normalization, clipping ratios, and advantage estimation in ways that are easy to overlook [20]. For Transformer-based RL specifically, the challenges are compounded: pre-normalization placement, the scale of value function targets, and gradient magnitude all interact in ways that differ from standard recurrent architectures [11,38–40]. Existing UAV and swarm control work typically applies PPO or MAPPO with default hyperparameters and does not design for the long-horizon sequence regime. Our steady-state mechanism—combining pre-normalized Transformer blocks with dynamic reward scaling and adaptive gradient clipping—treats training stability as a first-class design requirement rather than a tuning afterthought. Broader work on the robustness of learned representations under adversarial signal-level manipulation [41] and on post hoc refinement when sequential predictors miss structural shifts [42] reinforces a common theme: learning pipelines that process time-series or sensory streams must remain stable under perturbation and distribution shift—concerns that parallel our objective for embedded MARL policies under jamming-induced non-stationarity. In parallel to learning stability at the decision-making layer, the UAV literature also develops model-based reliability mechanisms such as actuator fault detection and isolation (FDI) for quadrotors modeled as LPV systems [43]. These approaches address low-level safety and diagnosis, whereas our focus is on high-level multi-agent decision making under awareness loss; they are complementary for real deployments.

3. Problem Formulation

3.1. Simulation Scenario and Entity Modeling

To evaluate CIDA’s algorithmic contributions under conditions that simultaneously activate both target failure modes (fragmented perception and motivation collapse), we construct a simulation benchmark in which a UAV swarm must navigate through a contested

area, suppress or bypass layered threat zones under strong electromagnetic jamming, and neutralize the target facility. The simulation environment is not intended to model any specific real-world system; it is deliberately designed to stress-test the algorithmic components of CIDA in the most demanding configuration. The terrain is a canyon-like environment rendered in a simulation platform; its physical and sensor boundaries are projected onto a 2D Euclidean plane $\mathcal{D} \subset \mathbb{R}^2$ of size $L \times W = 40 \times 40 \text{ km}^2$ for RL training (Figure 1). This 2D abstraction preserves the horizontal geometry of threat zones and mission waypoints while intentionally omitting line-of-sight (LOS) effects induced by terrain elevation. In the current benchmark, engagement and sensing ranges are modeled as isotropic disks in the horizontal plane, i.e., they depend on planar distance only; terrain-induced shadowing would require a 3D visibility/raycast model (or a learned occlusion model) to determine whether an actor is actually detectable or engageable at a given location. We did not include such coupling because of two independent considerations. First, *operational-geometry consistency*: in the adopted scenario, the adversarial air-defense posture already minimizes terrain-induced shadowing by design. Long-range search radars and SAM systems are emplaced on the canyon's flanking ridges to maximize their detection and engagement envelopes—these sensors therefore already clear the canyon crest; so, their horizontal-plane disks are faithful to their true coverage. AAA batteries, which serve as short-range terminal defense, are sited on the valley floor, where their local role is unaffected by the same crestline occlusion. The cooperative UAV swarm, moreover, operates at a cruise altitude on the order of several kilometers, whereas the canyon's relief is on the order of hundreds of meters; from this high-grazing-angle viewpoint, the projected “terrain shadow” area is small relative to the engagement footprint. Projecting the battlefield onto a 2D horizontal plane is therefore a small-error approximation under the assumptions of this scenario and does not materially distort the effective radar/SAM coverage or the reachable free space. Second, *algorithmic focus*: our goal in this paper is to isolate the algorithmic question of *cognitive belief reconstruction under communication denial* and *intent-driven motivation*, rather than to conflate it with platform-specific sensor/weapon LOS physics. We clarify this limitation and note that extending the environment to terrain-aware anisotropic engagement zones—replacing the isotropic disks with visibility-constrained regions computed from a 3D height field—is a natural next step (Section 6).

The cooperative agent group is a homogeneous swarm of $N = 5$ mission-capable UAVs, $\mathcal{U} = \{u_1, \dots, u_N\}$. The threat environment is $\mathcal{E} = \Gamma \cup \mathcal{K} \cup \mathcal{H}$: the target facility $\Gamma = \{e_{cp}\}$ (rendered as a command post in the simulation) is the mission objective; the active-denial layer $\mathcal{K} = \{e_{jam}, e_{sam,1:4}\}$ comprises a jammer radar and four SAM sites; and the supplementary layer $\mathcal{H} = \{e_{aaa,1:2}, e_{fuel,1:2}\}$ contains AAA batteries and decoy targets designed to exhaust swarm resources.

3.2. Decision Process as a POMDP

Jamming-induced sensing loss and intermittent communication mean that no agent can observe the full global state. We therefore model the swarm's decision problem as a decentralized POMDP $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, \gamma \rangle$. The global state $s_t \in \mathcal{S}$ contains the positions, velocities, and operational status of all cooperative and adversarial entities in the simulation. The joint action space is $\mathcal{A} = \prod_{i=1}^N \mathcal{A}_i$, where each $\mathcal{A}_i$ is hierarchically structured into three decision layers:

- **Task** $a^{\text{task}} \in \{0, \dots, 10\}$: select a target to attack or hold idle (*what to do*).
- **Maneuver** $a^{\text{mov}} \in \{0, \dots, 63\}$: select a waypoint on an 8×8 grid (*where to go*).
- **Speed** $a^{\text{spd}} \in \{0, \dots, 6\}$: discrete velocity from 0 to 150 m/s (*how fast*).

State transitions $P(s_{t+1} \mid s_t, a_t)$ are induced by UAV kinematics and the stochastic engagement logic of the threat environment (parameters in Table 1).

Table 1. Key parameters of simulated entities.

Side	Entity	#	Speed	Sensing	Range	Ammo	Damage	Hit Rate
Cooperative	UAV	5	0–150 m/s	11 km (no jam)/9.7 km (jam)	8 km	10	35	90%
Adversarial	Target facility	1	0	N/A	N/A	N/A	N/A	N/A
Adversarial	Jammer radar	1	0	Global	N/A	N/A	N/A	N/A
Adversarial	SAM site	4	0	6 km	5 km	4	51	60%
Adversarial	AAA battery	2	0	3 km	2.5 km	20	25	15%
Adversarial	Decoy fuel depot	2	0	N/A	N/A	N/A	N/A	N/A

The observation available to agent i depends on whether the jammer is active. Let $I_{jam}(t) \in \{0, 1\}$ indicate jammer status; the observation is

$$o_{i,t} = \begin{cases} o_{i,t}^{local}, & \text{if } I_{jam}(t) = 1, \\ o_{i,t}^{local} \cup \bigcup_{j \neq i} o_{j,t}^{local}, & \text{if } I_{jam}(t) = 0, \end{cases} \quad (1)$$

where $o_{i,t}^{local}$ contains all entities within the sensing radius R_{det} . When the jammer is active, each agent sees only its own local sensor cone, and swarm-wide situational awareness is effectively denied. However, physical communication loss does not equate to complete ignorance: the cognitive stream’s Transformer retains access to the full observation history $\tau_{i,t}$ and can re-attend to prior detections—for example, the last known position of a SAM site—thereby maintaining a working threat belief even when no new information is available. This temporal compensation is the key mechanism that breaks the symmetry between jamming-active and jamming-inactive regimes in the policy’s behavior.

In the formal POMDP tuple, the reward function R corresponds solely to the environment’s extrinsic feedback r_t^{ext} , which reflects mission milestones: successfully striking a target, UAV loss, mission completion, and episode timeout. CIDA augments this environmental signal with a dense intrinsic shaping term r_t^{int} produced by the intent stream—a design choice external to the POMDP definition, which leaves the environment unchanged. The learning objective over the augmented signal is

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t (r_t^{ext} + \beta r_t^{int}) \right], \quad (2)$$

where β weights the shaping signal, and $\gamma \in [0, 1)$ is the discount factor.

The extrinsic feedback $r_t^{ext} = R_{base}^t$ is the common baseline reward used by *all* methods compared in this study (CIDA and every baseline/ablation group in Section 5); it is therefore part of the environment rather than of CIDA, and we specify it explicitly here for reproducibility. It decomposes into four additive modules—survival/efficiency, self-protection, task execution, and formation safety:

$$R_{base}^t = \underbrace{c_{alive} N_{alive,t} + c_{step}}_{\text{survival/efficiency}} + \underbrace{c_{hp} \Delta H P_{lost,t} + c_{crash} N_{crash,t}}_{\text{self-protection}} + \underbrace{c_{dmg} \Delta H P_{dealt,t}^{tgt}}_{\text{task execution}} + \underbrace{R_{kill,t} + c_{col} N_{col,t}}_{\text{formation safety}}. \quad (3)$$

Here, $N_{alive,t}$ is the number of surviving UAVs at step t (scaled by c_{alive} , a per-agent survival incentive), and c_{step} is a small per-step time penalty that discourages indefinite loitering; $\Delta H P_{lost,t}$ is the total HP lost by the cooperative side in this step (penalized by c_{hp}), and $N_{crash,t}$ is the number of UAVs destroyed this step (penalized heavily by c_{crash}); $\Delta H P_{dealt,t}^{tgt}$ is the damage dealt to adversarial entities (rewarded by c_{dmg}), and $R_{kill,t}$ is the lump-sum

kill reward obtained when an adversarial entity is fully destroyed, with a per-entity value that reflects the strategic importance of that class (specified in Table 2); $N_{col,t}$ counts pairs of friendly UAVs whose separation falls below a 500 m safety threshold (penalized by c_{col}). All numerical values of the coefficients $\{c_{alive}, c_{step}, c_{hp}, c_{crash}, c_{dmg}, R_{kill,t}, c_{col}\}$ are listed in Table 2. The design of r_t^{int} is described in Section 4.3.

Table 2. Network architecture, baseline environment reward coefficients (Equation (3)), and CIDA-specific intent-stream reference weights for reproducibility. The rows in the “Baseline environment reward” block define $r_t^{ext} = R_{base}^t$ and are identical across all methods (CIDA and all baselines/ablations); the rows in the “CIDA intent-stream reference scale” block apply only to CIDA’s Equation (6).

Item	Value
<i>Network architecture (CIDA)</i>	
Transformer encoder layers (L)	4
Transformer model width (d_{model})	128
Attention heads	4
FFN hidden dimension	256
Embedding activation $\sigma(\cdot)$	ReLU
Feature projection (“Linear 512”)	128 $\rightarrow$ 512
Actor head MLP (each head)	2 layers, 512 hidden
Critic MLP	2 layers, 512 hidden
Total parameters of the execution network	≈ 0.366 M
<i>Baseline environment reward R_{base}^t (Equation (3)); shared by all methods</i>	
Survival incentive c_{alive}	+0.1
Per-step time penalty c_{step}	−0.5
HP-loss penalty c_{hp}	−1.5
UAV-crash penalty c_{crash}	−200
Damage-dealt reward c_{dmg}	+3
Kill reward $R_{kill,t}$: target facility (command post)	+28,000
Kill reward $R_{kill,t}$: SAM missile site	+2000
Kill reward $R_{kill,t}$: AAA battery	+1000
Kill reward $R_{kill,t}$: jammer radar/decoy fuel depot	+300
Formation-safety (collision) penalty c_{col}	−2
Collision safety radius	500 m
<i>CIDA intent-stream reference scale $v_{base}(e_j)$ (Equation (6)); CIDA only</i>	
$v_{base}(e_j)$: SAM site	1.0
$v_{base}(e_j)$: AAA battery	0.5
$v_{base}(e_j)$: jammer radar	0.8
$v_{base}(e_j)$: decoy fuel depot	0.2
$v_{base}(e_j)$: target facility (mission)	5.0
Geometry coefficient k_{geo}	1.0

4. Cognitive–Intent Dual-Stream Architecture (CIDA)

4.1. Overview

Two failure modes drive the design of CIDA. The first is *fragmented perception*: under jamming, agents lose shared situational awareness and must infer the current threat configuration from their own local history alone. The second is *motivation collapse*: reward signals dominated by survival pressure push the policy toward passive avoidance, suppressing the forward-approach behavior needed to complete the mission. Addressing one without the other is insufficient—a well-informed agent with no reason to advance will loiter; an aggressive agent with no situational awareness will make reckless self-defeating commitments.

CIDA's response is *orthogonal decoupling at the representation level*. A dedicated cognitive stream reconstructs threat belief from observation history, giving each agent an internally consistent estimate of environment state. A dedicated intent stream translates mission priorities into dense reward gradients, ensuring the optimization has a persistent drive toward the strategic objective. The two streams address different aspects of the control problem—the cognitive stream operates as a *state representation module* (what is the current situation?), while the intent stream operates as a *reward shaping module* (what is worth doing?)—and neither places constraints on the other during learning. Critically, they are *functionally coupled* during policy gradient updates: the belief feature $b_{i,t}$ produced by the cognitive stream serves as the input to the actor; so, intent-shaped reward gradients flow back through the same representation. This coupling is what enables mutual reinforcement rather than mere co-existence. Both streams feed the same actor–critic policy (Figure 2).

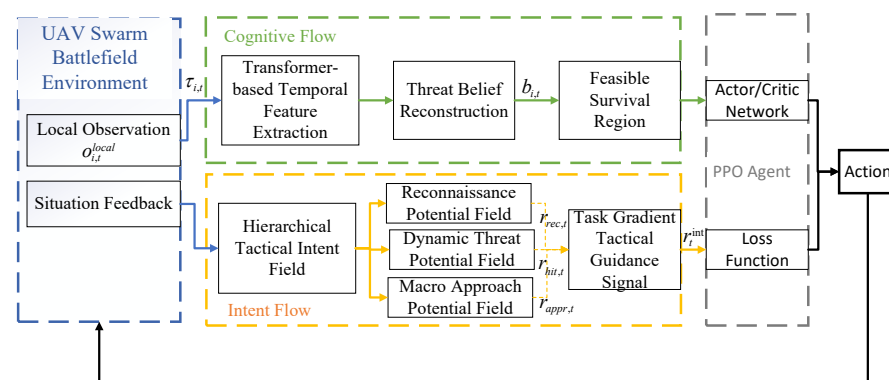


Figure 2. Overview of the proposed Cognitive–Intent Dual-Stream Architecture (CIDA). The cognitive stream reconstructs threat belief and a survivability-feasible region from local observation histories via Transformer-based temporal feature extraction; the intent stream encodes mission intent as a hierarchical tactical intent field (reconnaissance potential, dynamic threat potential, and macro approach potential) that provides dense guidance gradients. The two streams are decoupled at the representation level yet fused through a shared actor–critic PPO framework. Local observations $o_{i,t}^{local}$, the K -step history window $\tau_{i,t}$, the belief feature $b_{i,t}$ produced by the cognitive stream, the three intrinsic reward components r_{rec} , r_{hit} , r_{appr} , and the fusion point at which extrinsic and intrinsic rewards enter the PPO update are labelled explicitly with bolded arrows and symbol annotations.

4.2. Cognitive Stream: Transformer-Based Belief Reconstruction

For agent u_i , we concatenate the K most recent observations into a history window $\tau_{i,t} = [o_{i,t-K+1}, \dots, o_{i,t}] \in \mathbb{R}^{K \times d_{obs}}$. Each observation is projected to a latent embedding, normalized, and augmented with positional encoding:

$$E_{i,t} = \text{PE}(\sigma(\text{LN}(\tau_{i,t}W_e + b_e))). \quad (4)$$

Here, $W_e \in \mathbb{R}^{d_{obs} \times d_{model}}$ and $b_e \in \mathbb{R}^{d_{model}}$ are the learnable embedding projection parameters, d_{model} is the Transformer model width, LN is layer normalization, and $\sigma(\cdot)$ denotes the element-wise ReLU($\cdot$) activation. The sequence is then processed by a pre-normalized Transformer encoder. Placing layer normalization *before* the attention and feed-forward sub-layers (Pre-LN [40]) is critical: it keeps the residual stream well-scaled throughout training and is the primary structural choice that makes the Transformer tractable for on-policy RL:

$$\begin{aligned} H'_l &= H_{l-1} + \text{MSA}(\text{LN}(H_{l-1})), \\ H_l &= H'_l + \text{FFN}(\text{LN}(H'_l)). \end{aligned} \quad (5)$$

We set the encoder input as $H_0 = E_{i,t}$ to connect the embedding in the previous equation to the Transformer recurrence. The last-token representation $b_{i,t}$ serves as the agent's belief feature. It summarizes the observation history into a fixed-length vector that captures both the current local view and whatever can be inferred about the recent past—the closest approximation to a belief state that is feasible under communication denial.

The choice of Transformer over a recurrent architecture (e.g., GRU or LSTM) is deliberate. RNNs are prone to vanishing gradients over long sequences; in our setting, a jammer can suppress informative observations for many consecutive steps, and an RNN's hidden state tends to dilute those distant detections before they can influence the current decision. The Transformer's self-attention mechanism has no such temporal decay: it can directly re-attend to any step in the history window—for instance, the last confirmed SAM position before communication was lost—regardless of how many uninformative steps have elapsed since. This random-access recall is precisely the property needed when the most diagnostic observations may be arbitrarily old.

4.3. Intent Stream: Hierarchical Tactical Intent Field

Without a persistent incentive to advance, a well-trained agent may learn to survive indefinitely while never completing the mission—the passive-avoidance trap. The intent stream counters this by decomposing mission objectives into a hierarchy of three dense reward components, each targeting a different timescale of behavior.

Reconnaissance potential rewards the first detection of each adversarial entity, encouraging agents to actively explore rather than remain within already-mapped zones:

$$r_{rec,t} = \eta_{rec} \sum_{e_j \in \mathcal{E}} \mathbb{I}(I_{det}(e_j, t) \wedge \neg I_{det}(e_j, t-1)). \quad (6)$$

Dynamic threat potential weights target-neutralization rewards by each entity's strategic proximity to the target facility; so, the policy learns to prioritize threats that block the approach corridor over peripheral assets:

$$r_{hit,t} = \sum_{e_j \in \mathcal{E}_{hit}} v_{base}(e_j) \left(1 + k_{geo} \frac{d(e_j, e_{cp})}{W + \epsilon} \right). \quad (7)$$

In our implementation, $\epsilon = 10^{-6}$ is a numerical stability constant. The scalar $v_{base}(e_j)$ is a dimensionless CIDA-specific reference scale for the intent stream's spatial shaping—not the environment's per-entity kill reward $R_{kill,t}(e_j)$ in Equation (3), which is common to all methods. We set $v_{base}(e_j)$ to a small fixed class weight (values in Table 2) so that the geometry-aware factor $(1 + k_{geo} d(e_j, e_{cp}) / (W + \epsilon))$, rather than the scale of v_{base} , determines the shape of the intent gradient; the overall magnitude of the intent signal is then further controlled by the global weight β in Equation (2). The nonlinear scaling factor $\left(1 + k_{geo} \frac{d(e_j, e_{cp})}{W + \epsilon}\right)$ is specifically designed to prevent *aggressive myopia*—the tendency for an intent-only agent to commit to neutralizing the nearest or most accessible target regardless of whether doing so advances the mission. Entities far from the target facility receive a higher base multiplier, because neutralizing them yields a proportionally larger reward boost when the swarm subsequently closes distance. As the swarm advances deeper and $d(e_j, e_{cp})$ shrinks for the remaining threats, the gradient steepens, reinforcing commitment to the final approach once the outer threat zones have been cleared.

Macro approach potential rewards each step in which the swarm closes distance to the target facility, providing a global pull that sustains forward momentum throughout the episode:

$$r_{appr,t} = \eta_{appr} [D_{t-1}(\mathcal{U}, e_{cp}) - D_t(\mathcal{U}, e_{cp})]. \quad (8)$$

The composite shaping signal $r_t^{int} = r_{rec,t} + r_{hit,t} + r_{appr,t}$ spans short-horizon exploration, mid-horizon threat suppression, and long-horizon approach, giving the intent stream a multi-resolution grip on tactical behavior. The total reward used for training is $r_t = r_t^{ext} + \beta r_t^{int}$.

4.4. Dual-Stream Fusion with PPO

The two streams are coupled through a shared actor–critic network (Figure 3). Before entering the policy heads, the belief feature $b_{i,t}$ passes through a *Feature Amplification Layer*—a gated element-wise scaling block that re-weights the belief dimensions by their estimated task relevance, followed by a fully connected projection to a 512-dimensional hidden representation (*Linear 512*). This two-stage preprocessing ensures that task-relevant features in $b_{i,t}$ are amplified before the gradient from the intent-shaped reward reaches the cognitive stream, improving the quality of the credit signal flowing back to the Transformer. The actor then takes this 512-dimensional vector and outputs factorized distributions over the three action heads:

$$\pi_{\theta}(a_t | b_t) = \pi_{task}(a^{task} | b_t) \pi_x(a^x | b_t) \pi_y(a^y | b_t) \pi_{spd}(a^{spd} | b_t). \quad (9)$$

The critic estimates $V_{\phi}(b_t)$ from the same representation. Advantages are computed via GAE [36]:

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l (r_{t+l} + \gamma V_{\phi}(b_{t+l+1}) - V_{\phi}(b_{t+l})). \quad (10)$$

With importance ratio $\rho_t(\theta) = \pi_{\theta}(a_t | b_t) / \pi_{\theta_{old}}(a_t | b_t)$, the PPO objective with value and entropy regularization is

$$\mathcal{J}_{PPO}(\theta, \phi) = \mathbb{E}[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t)] - c_v \mathbb{E}[(V_{\phi}(b_t) - \hat{R}_t)^2] + c_e \mathbb{E}[\mathcal{H}(\pi_{\theta}(\cdot | b_t))], \quad (11)$$

where $\mathcal{J}_{PPO}$ is maximized with respect to θ (policy parameters), and ϕ appears through the value term, $\hat{R}_t = \sum_l \gamma^l r_{t+l}$, and $\mathcal{H}$ is policy entropy. The intent stream's reward r_t^{int} enters through the advantage and value targets; so, its gradients propagate to both the actor and the critic without requiring a separate optimization step. This joint optimization is what makes the two streams mutually reinforcing rather than independent.

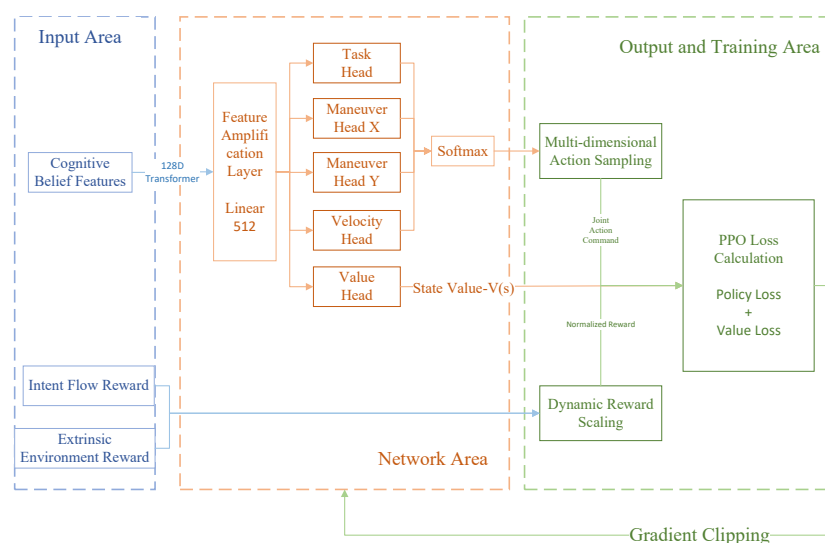


Figure 3. Multi-head PPO network with dual-stream fusion. Belief features from the cognitive stream parameterize the factorized actor heads (task, maneuver X/Y, speed), while the critic learns $V(b)$ from the same representation. Intrinsic intent shaping and extrinsic environment rewards are combined for advantage/value estimation; training is stabilized via dynamic reward scaling and gradient clipping.

4.5. Steady-State Training Mechanisms

Two practical problems arise when training a Transformer-based policy with PPO on long-horizon multi-agent episodes—problems that are specific to the Transformer architecture and do not appear with the same severity in standard MLP or GRU policies. First, the combined reward $r_t^{ext} + \beta r_t^{int}$ has a scale that shifts over training, as the policy and the jamming topology co-evolve, causing the value function to continually re-scale and introducing instability in the advantage estimates. Second, the Transformer’s gradient norms can spike sharply at transition points where the attention pattern reorganizes—a consequence of the self-attention mechanism’s quadratic interaction structure—occasionally pushing the policy out of the PPO clipping region before the constraint can act. Standard gradient clipping with a fixed norm threshold provides insufficient protection because the spike magnitude itself changes as the attention pattern matures. These two failure modes together constitute the *Transformer divergence problem in RL*: when both occur simultaneously, the critic loses calibration, and the policy collapses abruptly, as confirmed empirically in Figure 5.

We address both with a steady-state mechanism. *Dynamic reward scaling* normalizes returns by a running estimate of their standard deviation, keeping the value function learning rate effectively constant:

$$r_t^{norm} = \frac{r_t^{ext} + \beta r_t^{int}}{\sigma_{R,t} + \epsilon}. \quad (12)$$

Adaptive gradient clipping caps the gradient norm before each parameter update, suppressing the occasional large steps that cause abrupt policy change:

$$\nabla_{\theta} \leftarrow \nabla_{\theta} \cdot \min\left(1, \frac{\delta_{clip}}{\|\nabla_{\theta}\|_2}\right), \quad (13)$$

where $\delta_{clip} = 0.5$ in our experiments. Together with Pre-LN, these two mechanisms form a layered defense that directly resolves the Transformer divergence problem: dynamic reward scaling prevents value-function drift, and adaptive gradient clipping suppresses the attention-reorganization spikes that standard fixed-norm clipping cannot catch. The combined effect—smooth critic loss and monotone success-rate convergence—is confirmed by the ablation shown in Figure 5.

5. Experiments

5.1. Environment and Parameters

Simulation runs were conducted in a 3D continuous simulation environment; the altitude is fixed at a cruise height so that decisions and threat interactions are captured in the 2D horizontal plane described in Section 3. An episode succeeds when the swarm neutralizes the target facility, and it fails if all UAVs are destroyed or 1000 steps elapse. As long as the jammer radar is operational, each agent’s sensing range is reduced from 11 km to 9.7 km, and inter-UAV communication is fully denied. The key entity parameters are listed in Table 1; the training hyperparameters are summarized in Table 3. To assess whether the observed performance gap generalizes beyond the single canyon-like configuration in Figure 1, we constructed a second 2D region configuration with a qualitatively different free-space layout and threat placement (Figure 4). In contrast to the narrow canyon corridor, this scenario features an *open-terrain distributed-defense* layout with a much wider operational front: four SAM sites form a distributed mutually overlapping outer cordon of overlapping engagement zones; the jamming radar is placed deep in the rear and protected by the SAM umbrella; and two AAA batteries provide a dense short-range inner ring around a centrally located command post. The absence of natural canyon walls and the mutual overlap of the SAM engagement disks eliminate any “safe corridor” of free space; so, the swarm can

no longer bypass the outer defense layer the way it does in the canyon configuration—a qualitatively different tactical problem.

Table 3. Training hyperparameters.

Hyperparameter	Value
Learning rate	1×10^{-4}
Discount factor γ	0.9
GAE λ	0.95
PPO clipping ϵ	0.2
Reward normalization stability constant ϵ	10^{-6}
Intent shaping weight β	1.0
Entropy coefficient	0.01
Batch size	2048
Total training episodes	13,000
Observation window length K	64
Gradient clipping norm	0.5
Random seeds (per curve)	3

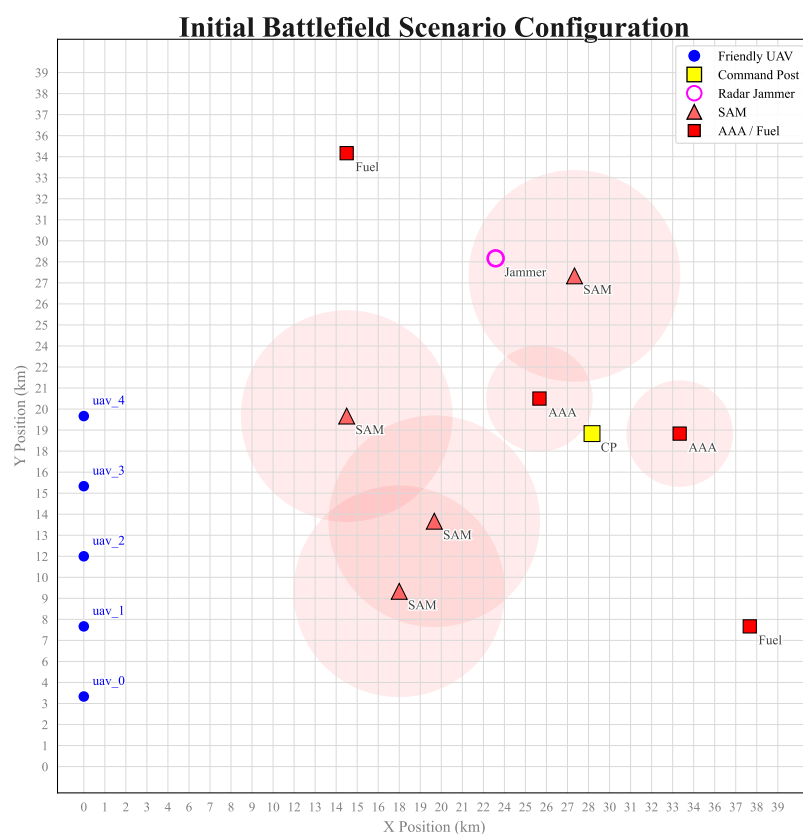


Figure 4. Second 2D region configuration used for the generalization study: an open-terrain distributed-defense scenario. Four SAM sites form a mutually overlapping outer cordon (red zones), the jamming radar is placed deep in the rear under SAM protection, and two AAA batteries (orange zones) form a short-range inner ring around the centrally located command post (target facility). Cooperative UAVs are initialized on the left.

We trained the full CIDA (Group-D) model on this new configuration using the same hyperparameters as in Table 3 (three seeds) and report the same five performance metrics as in Table 4. The aggregate results are summarized in Table 5. CIDA maintains a high task success rate (90.94%) and near-perfect swarm survivability (99.09%) on the open-terrain layout, a drop of only ~ 5 percentage points from the canyon scenario. Because the overlapping SAM cordon blocks all traversal corridors, the swarm now *deliberately* elects to

suppress the outer SAM layer rather than bypass it, and the SAM neutralization rate rises to 74.98%; the jammer neutralization rate nonetheless remains very low (6.66%), indicating that—just as in the canyon configuration—CIDA agents continue to rely on the cognitive stream’s belief reconstruction and do not perform high-risk low-value attacks on the jammer to restore communication. Taken together, this second configuration confirms that the behaviors induced by CIDA—high success, high survivability, adaptive selection between *bypass* and *breach*, and bypass of the jammer—are not artefacts of the specific canyon geometry but reflect a robust algorithmic pattern that generalizes across qualitatively different free-space/obstacle layouts.

Table 4. Final performance statistics (mean $\pm$ std, %). Best result per metric in **bold**. [†] Proposed method.

Metric	Group-A	Group-B	Group-C	Group-D (CIDA) [†]
Task success rate	12.21 $\pm$ 3.28	54.00 $\pm$ 0.49	57.33 $\pm$ 11.65	96.15 $\pm$ 0.10
Swarm survivability	96.17 $\pm$ 0.22	86.09 $\pm$ 3.14	99.97 $\pm$ 0.02	99.97 $\pm$ 0.03
SAM neutralization rate	28.27 $\pm$ 1.46	30.74 $\pm$ 2.29	30.45 $\pm$ 2.50	26.38 $\pm$ 0.42
AAA neutralization rate	40.07 $\pm$ 3.16	33.58 $\pm$ 2.46	49.84 $\pm$ 0.22	49.97 $\pm$ 0.05
Jammer neutralization rate	13.14 $\pm$ 3.86	26.70 $\pm$ 6.30	20.24 $\pm$ 7.11	4.81 $\pm$ 1.43

Table 5. Generalization performance of CIDA (Group-D) on the second open-terrain distributed-defense configuration shown in Figure 4 (mean $\pm$ std, %, three seeds).

Metric	CIDA (Group-D) on the Open-Terrain Layout
Task success rate	90.94 $\pm$ 2.05
Swarm survivability	99.09 $\pm$ 0.08
SAM neutralization rate	74.98 $\pm$ 0.09
AAA neutralization rate	49.87 $\pm$ 0.23
Jammer neutralization rate	6.66 $\pm$ 2.25

Here “Damage” is the per-hit damage applied to an entity’s hit points (HP). We set UAV HP to 100 and SAM HP to 100 (AAA HP to 60, jammer HP to 80) so that the expected time-to-kill is consistent with the engagement parameters; thus, e.g., a SAM hit (damage 51) typically requires two successful hits to destroy a UAV.

5.2. Baselines, Ablations, and Metrics

We design a stepwise evaluation protocol comprising two external benchmarks and three internal ablations. The protocol is structured to answer three questions: (i) How difficult is the task without any memory or intent (Group-A, MAPPO+RNN)? (ii) What does each stream contribute in isolation (Group-B, Group-C)? (iii) Does the dual-stream combination produce a nonlinear gain beyond either component alone (Group-D)?

Groups A and B use a memoryless MLP encoder; Groups C and D use a pre-normalized Transformer encoder with a history window of $K = 64$ steps; MAPPO+RNN uses a GRU recurrent encoder. Table 6 summarizes the remaining configuration differences and the specific hypothesis each group is designed to validate.

We report the following metrics (Table 4). **Task success rate** is the percentage of evaluation episodes in which at least one UAV reaches and neutralizes the target facility before timeout. **Swarm survivability** is the percentage of episodes in which at least one UAV remains operational at the end of the episode. **Neutralization rate** for an entity class is the percentage of episodes in which at least one entity of that class is destroyed (for SAM/AAA/decoy), and for the jammer, it is the percentage of episodes in which the jammer radar is destroyed. We report the task success rate, swarm survivability, and neutralization rates of key adversarial entities (jammer radar, SAM sites, AAA batteries, and decoys).

Table 6. Ablation group configurations and validation targets. [†] Proposed method.

Group	Cognitive Stream	Intent Stream	Training Mechanism	Validation Target
Group-A (Baseline)	×	×	Standard PPO	Quantify task difficulty; PPO+MLP performance floor
Group-B (Intent only)	×	✓	Standard PPO	Isolate the effect of tactical intent shaping
Group-C (Cognition only)	✓	×	Steady-state	Isolate the effect of belief reconstruction
Group-D (CIDA) [†]	✓	✓	Steady-state	Proposed full dual-stream model
MAPPO+RNN [37]	×	×	Standard MAPPO	Mainstream sequential-memory benchmark

5.3. Effect of Steady-State Training

Before comparing CIDA against baselines, we isolate the effect of the steady-state mechanism on Group-C (cognition-only), in order to demonstrate that it resolves a Transformer-specific divergence failure that cannot be addressed by simply tuning PPO hyperparameters. Figure 5 shows the critic loss and task success curves with and without stabilization. Without it, the critic loss oscillates for several thousand episodes and then collapses, dragging the success rate down with it—a clear policy collapse caused by the compounding of value-scale drift and attention-reorganization gradient spikes. With stabilization, the critic loss decreases monotonically from the start, and the success rate rises without catastrophic interruption. This confirms that the steady-state mechanism is not merely a refinement: it is a prerequisite for making Transformer-based MARL reliable in this scenario, solving the divergence problem at its root rather than tolerating it.

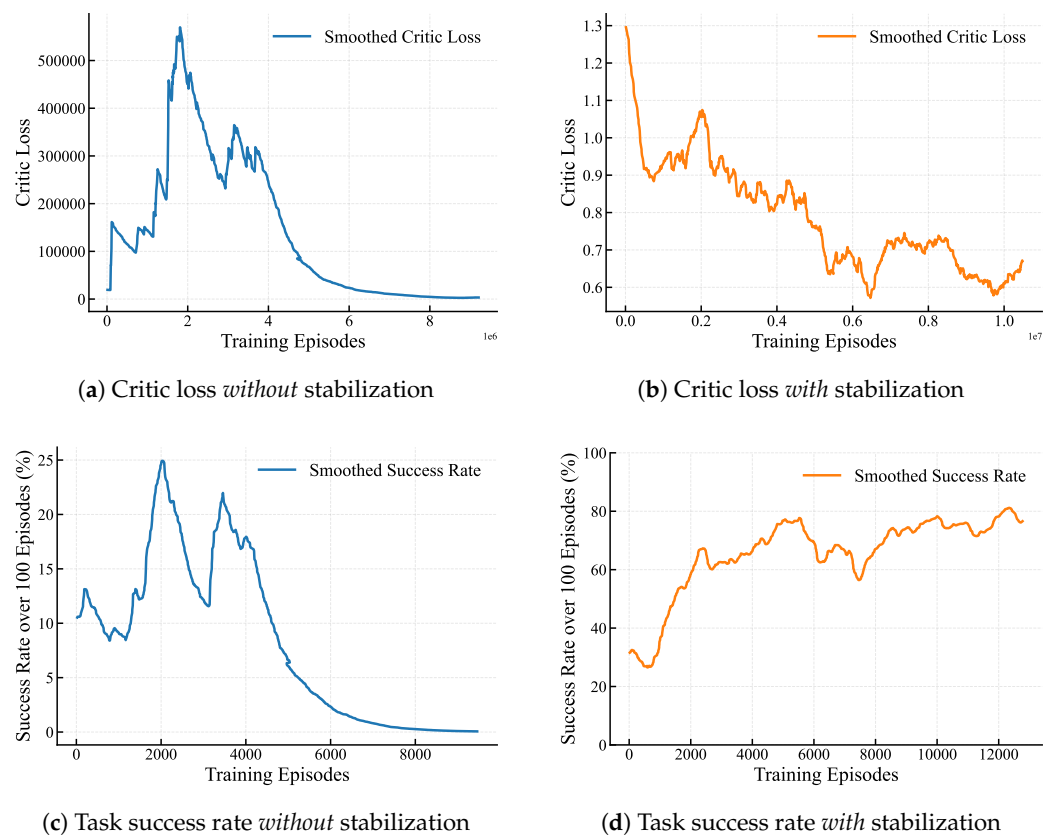


Figure 5. Steady-state training effect on Group-C (cognition-only stream). **Without** stabilization (a,c): the critic loss exhibits large oscillations followed by collapse, and the task success rate converges to a poor local optimum. **With** stabilization (b,d): critic loss decreases smoothly and remains bounded, while the success rate converges stably without catastrophic drops, confirming that the steady-state mechanism is essential for reliable long-horizon sequence learning.

5.4. Main Results and Analysis

5.4.1. Comparison to Mainstream Baselines

Figure 6 shows the success-rate learning curves for the three external comparisons. Group-A is revealing in its own right: despite 9000 training episodes, the memoryless baseline never exceeds 12%, confirming that the fog-of-war scenario is genuinely difficult and cannot be solved by a stronger policy optimizer alone. MAPPO+RNN learns more efficiently in early training, reaching $\sim 20\%$ by episode 2000, but then stalls at 25.28%. The curve flattens rather than continuing to rise, which suggests the recurrent architecture is hitting a behavioral ceiling rather than a capacity limit.

CIDA is qualitatively different. It rises steeply from the first thousand episodes and converges smoothly to 96.15%, with a variance band that narrows as training progresses. The gap relative to MAPPO+RNN—roughly 70 percentage points—is large enough that it is unlikely to close with additional training or hyperparameter tuning.

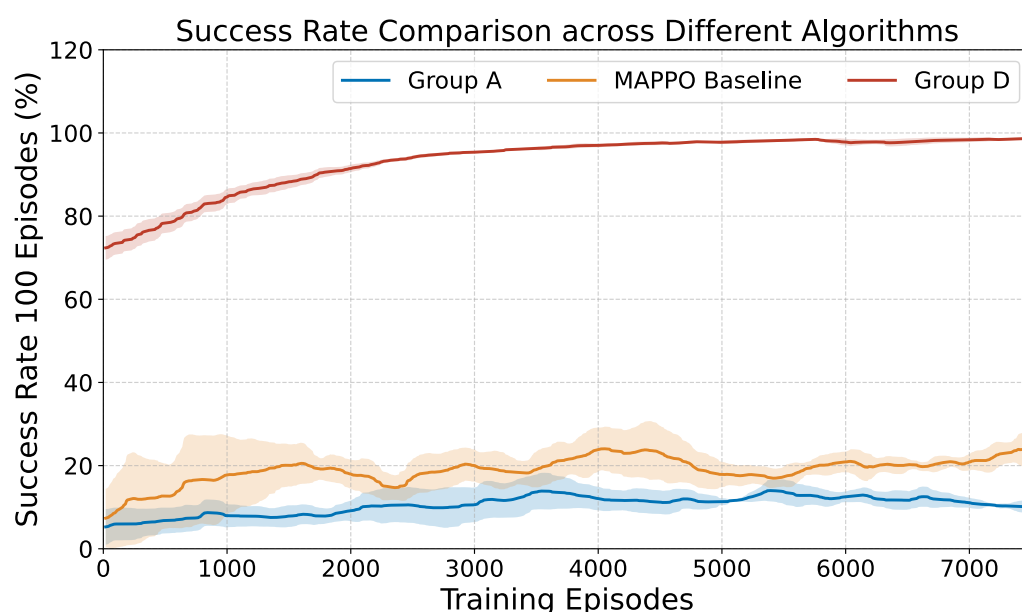


Figure 6. Task success rate learning curves for Group-A (memoryless PPO), MAPPO+RNN, and CIDA (Group-D). Shaded bands denote ± 1 standard deviation over three seeds. Group-A saturates near 12%, quantifying the intrinsic task difficulty. MAPPO+RNN improves early but plateaus at 25.28% after ~ 2000 episodes. CIDA converges smoothly to 96.15%.

Figure 7 shows why MAPPO+RNN stalls. The trajectory inspection reveals a consistent pattern: agents approach the outer threat perimeter, survive there, and never commit to an advance. They oscillate near the target facility's threat boundary with high survivability (99.98%) and moderate peripheral asset destruction, yet fail to advance the final distance to the objective. This is the passive-avoidance trap in its clearest form—the recurrent network has learned *where not to go* but has no internal drive to go anywhere in particular.

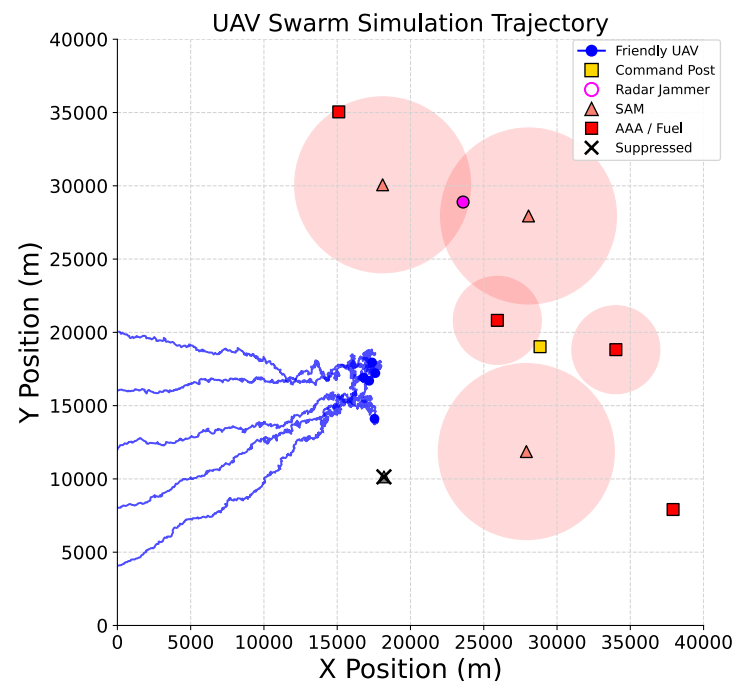


Figure 7. Typical failure trajectory of MAPPO+RNN in the fog-of-war scenario; the legend identifies UAV trajectories, threat engagement zones, the target facility, destroyed entities, and suppressed assets. Cooperative UAV trajectories cluster at the boundary of the outermost threat zone and exhibit persistent loitering near the target facility threat perimeter. Despite high swarm survivability (99.98%) and partial neutralization of peripheral SAM and AAA assets, the agents lack a decisive forward-approach drive and are trapped in a passive-avoidance local optimum—repeatedly oscillating near the threat perimeter without advancing toward the target facility, ultimately resulting in task failure.

5.4.2. Ablation Study and Coupling Gain

Figure 8 and Table 4 present the ablation results. The four groups exhibit markedly different performance trajectories, which we analyze in turn.

Group-A serves as the difficulty baseline. Without history integration, agents cannot distinguish active jamming from normal operation and fail to reconstruct the threat boundary. The success rate remains near 12% throughout all 13,000 training episodes, confirming the intrinsic difficulty of the fog-of-war scenario.

Group-B demonstrates the value and the limit of intent shaping. Dense guidance gradients yield approximately a four-fold improvement in early-episode success rates. However, intent guidance operating on raw fragmented observations cannot adapt to the dynamic jamming topology. The policy saturates at ~54%—agents are guided *where* to go but not *whether it is safe* to proceed.

Group-C demonstrates the value and the limit of belief reconstruction. The survivability improves dramatically (99.97%), and the success rate reaches ~57%, yet Group-C exhibits a characteristic instability: step-like oscillations and a substantially larger variance band (std 11.65% vs. 0.49% for Group-B). This arises because the Transformer’s self-attention mechanism is highly sensitive to policy distribution drift in on-policy RL [20]: momentary attention misalignment triggers sudden performance collapses that the steady-state mechanism mitigates but cannot fully eliminate. Crucially, without tactical intent the optimization lacks an explicit forward-approach drive, causing Group-C to converge to a *conservative-avoidance* strategy—reconstructing threat boundaries accurately yet failing to commit to the strategic objective.

Group-D resolves the complementary failure modes of Groups B and C. In early training, the intent field provides fast structured guidance, and the dual-stream reaches

~60% success within the first 1000 episodes. As training progresses, the cognitive stream breaks through the intent ceiling by enabling context-aware traversal decisions; simultaneously, the intent field's low-variance gradients smooth out the Transformer's oscillations. The result is a smooth learning curve converging to 96.15%—a *nonlinear coupling gain* that cannot be explained by simple addition of either stream's individual contribution.

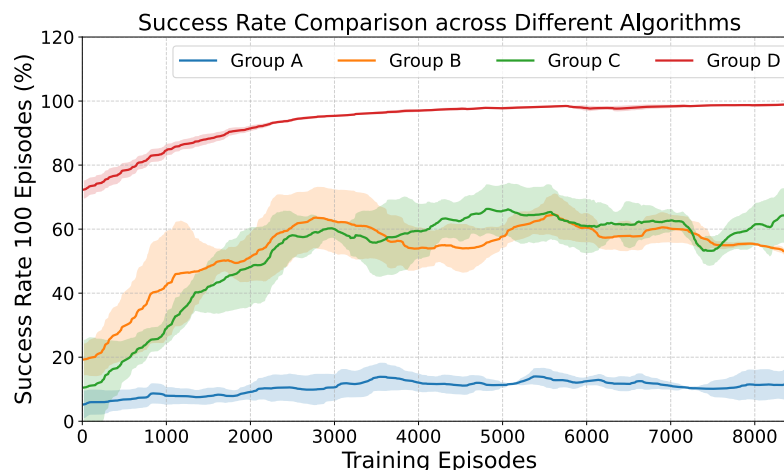


Figure 8. Task success rate learning curves across all ablation groups (A–D) over 13,000 training episodes. Shaded bands denote ± 1 standard deviation over three seeds. Group-A (memoryless PPO) remains near 12% throughout. Group-B (intent-only) rises quickly to ~54% but saturates early, limited by the absence of belief reconstruction. Group-C (cognition-only) achieves ~57% with higher variance, exhibiting conservative avoidance without tactical guidance. Group-D (full dual-stream CIDA) consistently outperforms all ablations and converges to 96.15%, demonstrating a nonlinear coupling gain that exceeds the sum of either stream alone.

Table 4 further reveals the multi-dimensional performance differences. In terms of swarm survivability, Group-B (intent-only) is the weakest at 86.09%, as aggressive intent shaping without belief reconstruction causes unnecessary exposure to defended areas. Group-C and Group-D both achieve near-perfect survivability (99.97%), confirming that the cognitive stream is the primary driver of safety. In Group-D, the intent field's tactical ladder guides agents to bypass high-threat SAM zones in favor of tactically weaker access routes; so, the SAM neutralization rate (26.38%) is actually lower than in Groups B and C—reflecting efficient goal-directed traversal rather than exhaustive engagement. The near-zero jammer neutralization rate in Group-D (4.81%) is particularly notable: unlike MAPPO+RNN and Group-B, which frequently target the jammer to restore communication, CIDA agents instead rely on their reconstructed belief state and bypass the jammer entirely, treating communication restoration as unnecessary once the threat knowledge is internalized.

5.4.3. Emergent Cooperative Tactics

Beyond the quantitative improvement, CIDA induces qualitatively different behavior that is worth examining. Figure 9 shows a representative late-training trajectory for Group-D. Two aspects stand out.

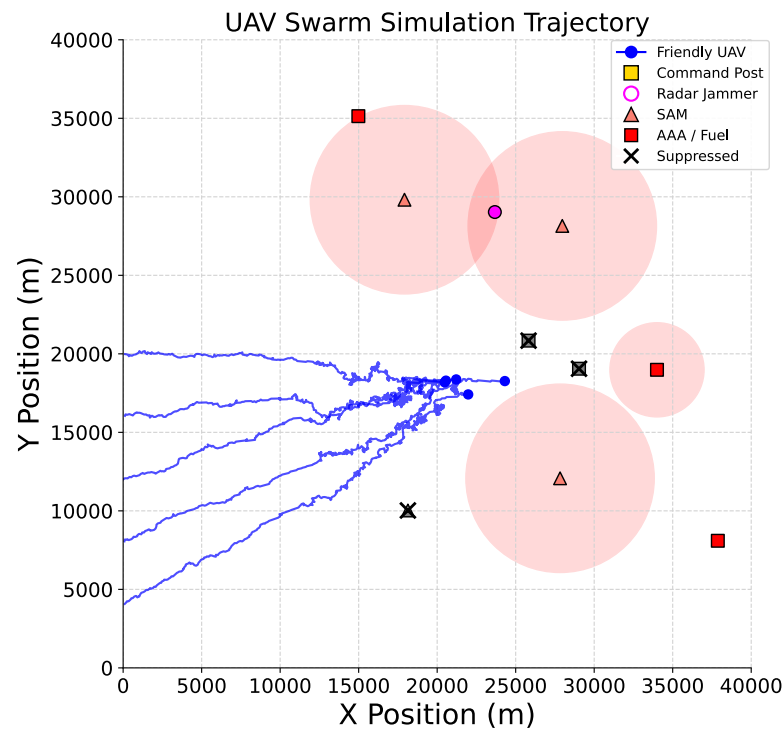


Figure 9. Representative emergent coordinated traversal tactic induced by CIDA (Group-D) in late training; the legend identifies the UAV trajectories, engagement zones, the target facility, and suppression markers. Rather than loitering at SAM engagement boundaries or blindly neutralizing the jammer radar to restore communication, the swarm leverages the cognitively reconstructed threat boundary to opportunistically suppress peripheral adversarial threat assets and execute a direct coordinated approach toward the strategically weak flank of the target facility. Crossed markers (×) indicate suppressed adversarial entities. This goal-directed behavior under partial observability demonstrates CIDA’s ability to overcome passive-avoidance traps and achieve high task success while maintaining swarm survivability.

First, the swarm does not attack the jammer radar. MAPPO+RNN agents frequently target the jammer to restore communication—a locally rational action whose mission value is indirect and delayed. CIDA agents leave the jammer alive and work from their reconstructed belief state instead. The jammer neutralization rate falls to 4.81% in Group-D, the lowest across all configurations. This is not a consequence of ignoring the jammer; it reflects a learned preference for the alternative. The cognitive stream provides enough situational awareness that restoring communication is simply no longer necessary.

Second, the swarm identifies and exploits the weakest threat sector. Rather than pressing into the densest part of the threat envelope or waiting at its boundary, agents converge on a route that threads between engagement zones, neutralizes a small number of blocking assets, and reaches the target facility directly. This coordinated behavior emerges from training alone, without any explicit route planning or inter-agent communication protocol beyond the shared observation when the jammer is inactive.

6. Discussion

The most striking result is the jump from roughly 12% (Group-A) to 96.15% (Group-D). A gap of this magnitude can appear implausible; so, it is worth explaining precisely why it arises. The key is that the task has two independent failure conditions, and existing methods resolve at most one of them. Without the cognitive stream, an agent operating under jamming cannot reliably distinguish safe corridors from threat zones; so, the safest learned behavior is to stay near the outer perimeter—Group-A’s 12% reflects not the limit

of PPO as an optimizer but the near-impossibility of the task when the agent is effectively blind to threat state. Without the intent stream, an agent with full belief reconstruction still has no reward incentive to accept the cost of the final approach—Group-C’s 57% with high variance reflects a policy that has learned the environment state accurately but has not learned to act on that knowledge aggressively. Group-D solves both simultaneously: the cognitive stream makes the risk assessable, and the intent stream makes it worth taking. The 96% figure is not a single large gain over a single baseline; it is the compounding of two independent fixes to two independent failure modes, which is why neither stream alone comes close.

The passive-avoidance trap is not a failure of learning capacity—MAPPO+RNN and Group-C both converge reliably—it is a failure of what the joint architecture asks the policy to do. Once both constraints are properly handled, the RL optimizer finds the solution with consistency (std 0.10%).

Intent shaping fixes the motivation problem but introduces a different vulnerability. Gradients computed from fragmentary observations can mislead the policy into aggressive commitments under conditions it cannot accurately assess, producing the opposite failure: high agent losses, lower survivability, and inconsistent success. Group-B’s 86% survivability rate, roughly 14 points below Group-C and Group-D, is a direct measure of this cost.

What CIDA provides is a clean separation between these two concerns. The cognitive stream’s job is epistemic—build an accurate enough belief about the current environment state to make the intent field’s gradients trustworthy. The intent stream’s job is motivational—ensure that the policy always has a reason to advance. When both are functioning, the combination is more than additive: the cognitive stream raises the ceiling of what intent can safely ask for, and the intent stream ensures that the cognitive stream’s representations are exercised across the full simulation environment rather than only the safe periphery.

A practical implication for system design is that training stability is not separable from architecture choice. The same Transformer that enables powerful belief reconstruction also introduces optimization fragility in the on-policy setting. Our steady-state mechanism addresses this, but the deeper lesson is that reliability engineering for RL-based autonomous systems deserves as much attention as the model architecture itself—particularly for safety-critical applications where policy collapse during deployment is unacceptable.

Several limitations bound the current conclusions. First, all experiments use a fixed configuration of five homogeneous UAVs against a static adversarial threat arrangement; performance in heterogeneous swarms where individual platforms differ in sensing radius, speed, and payload has not yet been evaluated. Second, the 2D horizontal-plane abstraction discards altitude dynamics that may be decisive in real canyon terrain, and extension to full six-DoF environments is a necessary future step before broader practical applicability can be claimed. Third, the jammer is modeled as a binary on/off switch; realistic adaptive jamming with directional and power-variable interference may require modifications to both the cognitive stream’s history encoding and the steady-state mechanism’s normalization schedule. These constraints limit direct transferability but do not undermine the validity of the orthogonal decoupling principle, which is scenario-agnostic. Fourth, terrain-aware line-of-sight is not modeled: in the current 2D projection, sensing/engagement radii are not occluded by terrain. A terrain-aware extension would replace disk ranges with visibility-constrained regions computed from the 3D height field; this would mainly affect (i) the observation function Ω (what can be detected) and (ii) the effective threat reach (what can engage). Importantly, CIDA itself does not assume isotropic ranges: the cognitive stream can ingest visibility-masked observations, and the intent stream can compute potentials

over any reachable/free-space map; we leave this integration as future work because it is orthogonal to the algorithmic focus of the present study.

Computational cost and deployability. At execution time, each UAV performs a single forward pass through the cognitive Transformer on a fixed window ($K = 64$) and a lightweight MLP actor; the full execution network has approximately 0.366 M parameters in total (Table 2). To quantify the runtime cost of this design, we benchmarked the per-step inference latency at batch size 1 (i.e., fully decentralized on-board execution, one forward pass per UAV per control tick) on two representative platforms: a consumer-grade laptop GPU (NVIDIA RTX 4060 Laptop) and a commercial off-the-shelf CPU that serves as a conservative proxy for embedded compute (Intel Core i7-14700HX, single-thread inference). The measured mean latency per forward pass is 1.869 ms on the RTX 4060 Laptop and 3.494 ms on the i7-14700HX CPU. Because typical on-board flight-control and tactical-decision loops on UAV platforms run at 10–50 Hz (a 20–100 ms budget per control cycle), a per-agent decision latency of ~ 3.5 ms on a CPU-only proxy leaves more than an order of magnitude of timing headroom, which is consistent with real-time on-board deployment and supports the claim that CIDA’s computational profile is compatible with embedded autonomy rather than limited to offline simulation.

Other applications of CIDA. The cognitive-intent decoupling principle is applicable beyond EW swarming whenever policies must act under incomplete information while avoiding passive reward hacking. Examples include multi-robot search-and-rescue with intermittent comms (cognition: belief over victim/obstacle map; intent: coverage and extraction potentials), maritime/ground convoy escort under sensor dropouts (cognition: threat belief; intent: formation maintenance and route-progress potentials), and warehouse fleets under dynamic occlusions (cognition: occupancy belief; intent: throughput and collision-avoidance potentials). We add these examples to broaden the context of CIDA’s relevance.

7. Conclusions

We introduced CIDA, a cognitive-intent decoupled reinforcement learning framework for cooperative UAV swarm control in contested environments. The core design principle—orthogonally decoupling belief reconstruction from tactical intent, each handled by a dedicated non-interfering stream—resolves the tension between survivability and mission completion that memory-only and intent-only approaches cannot overcome independently. The cognitive stream, built on a pre-normalized Transformer with a 64-step history window, gives each agent a working belief about the current threat configuration even when communication is denied. The intent stream provides a hierarchical reward field that keeps the optimization pointed at the mission objective across the full episode. A steady-state training mechanism that directly resolves the Transformer-specific gradient-divergence problem makes the joint optimization reliable in the non-stationary multi-agent setting.

In a complex terrain traversal scenario with four SAM sites, two AAA batteries, and a jammer radar, CIDA achieves 96.15% task success against 12.21% for memoryless PPO and 25.28% for MAPPO+RNN. Ablations confirm a nonlinear coupling gain, and behavioral analysis reveals qualitatively distinct tactics: CIDA agents bypass the jammer rather than attacking it and traverse through dynamically identified weak sectors rather than stalling at threat boundaries.

The orthogonal decoupling principle is not specific to the platform or the exact scenario studied here. Extending it to heterogeneous swarms with different sensor profiles, full six-DoF maneuvering, and domain-adaptive transfer to environments with realistic

communication delays—addressing the limitations identified in the Discussion—are the primary directions of ongoing work.

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DURC Statement: Current research is limited to the field of cooperative multi-agent reinforcement learning (MARL) and autonomous decision-making for robotic swarms, which is beneficial for enhancing the robustness and coordination of unmanned systems in complex partially observable environments (e.g., search-and-rescue or industrial monitoring under signal interference) and does not pose a threat to public health or national security. The authors acknowledge the dual-use potential of the research involving autonomous trajectory planning and cooperative control algorithms for unmanned systems in contested signal environments and confirm that all necessary precautions have been taken to prevent potential misuse. As an ethical responsibility, the authors strictly adhere to the relevant national and international laws about DURC. The authors advocate for responsible deployment, ethical considerations, regulatory compliance, and transparent reporting to mitigate misuse risks and foster beneficial outcomes.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AAA	Anti-Aircraft Artillery
CIDA	Cognitive-Intent Dual-Stream Architecture
DoF	Degrees of Freedom
FFN	Feed-Forward Network
GAE	Generalized Advantage Estimation
GRU	Gated Recurrent Unit
LN	Layer Normalization
MARL	Multi-Agent Reinforcement Learning
MAPPO	Multi-Agent Proximal Policy Optimization
MLP	Multi-Layer Perceptron
MSA	Multi-head Self-Attention
OODA	Observe-Orient-Decide-Act
PE	Positional Encoding
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
Pre-LN	Pre-Layer Normalization
RL	Reinforcement Learning
RNN	Recurrent Neural Network
SAM	Surface-to-Air Missile
UAV	Unmanned Aerial Vehicle

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